

Cells and Batteries

Inside a cell, a chemical reaction makes electricity — from Volta's first cell to today's rechargeable batteries.

How a cell makes electricity

A **cell** (or battery) makes electric current from a **chemical reaction** inside it.

The Voltaic (Galvanic) cell

It has two **electrodes** (rods of two **different** metals) dipped in a liquid called an **electrolyte** (a weak acid or salt solution). The reaction between the metals and electrolyte produces electricity. Current flows from the **positive** terminal through the circuit to the **negative** terminal. When the chemicals are used up, the cell is "dead".

Lemon cell (Activity 4.6): copper wire + iron nail in lemons (lemon juice = electrolyte) can light an LED. Salt solution works too.

Dry cell

Called "dry" because the electrolyte is a moist **paste**, not a liquid. A **zinc container** is the **negative** terminal; a central **carbon rod** with a metal cap is the **positive** terminal. It is a **single-use** cell.

Rechargeable batteries

Can be charged and reused many times (phones, laptops, inverters, vehicles), saving money and reducing waste — but they slowly wear out. Most common type today: the **lithium-ion** battery.

Volta vs Galvani: Galvani saw a frog's leg twitch with two metals and thought electricity came from the frog; Volta showed it came from the **metals + liquid**, and built the first battery.

Where students lose marks

Trap 1 — Electrodes must be two DIFFERENT metals. Same metals on both sides won't make a working cell.

Trap 2 — Electrolyte is a conducting liquid/paste. Pure water is a poor electrolyte; lemon juice or salt solution works. (In Fig 4.12, the lemon-juice cell lights the LED, the pure-water one does not.)

Trap 3 — "Dry" cell is not really dry. Its electrolyte is a moist paste; only the Voltaic cell uses a liquid electrolyte.

Dead cell: chemicals are used up, so it can't supply current; a dry cell is single-use, while rechargeables can be recharged.

Model answers — write them like this

Example A. Name the parts of a Voltaic cell and what makes the electricity.

Step 1: Two electrodes of different metals dipped in an electrolyte (weak acid/salt solution).

Step 2: A chemical reaction between metals and electrolyte produces current.

∴ Electrodes + electrolyte; the chemical reaction makes the electricity.

Example B. In Fig 4.12, why does the LED glow with lemon juice but not with pure water?

Step 1: Lemon juice is a good electrolyte; pure water is not.

Step 2: Only with a working electrolyte does the chemical reaction drive a current.

∴ The lemon-juice cell makes current and lights the LED; the pure-water one does not.

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